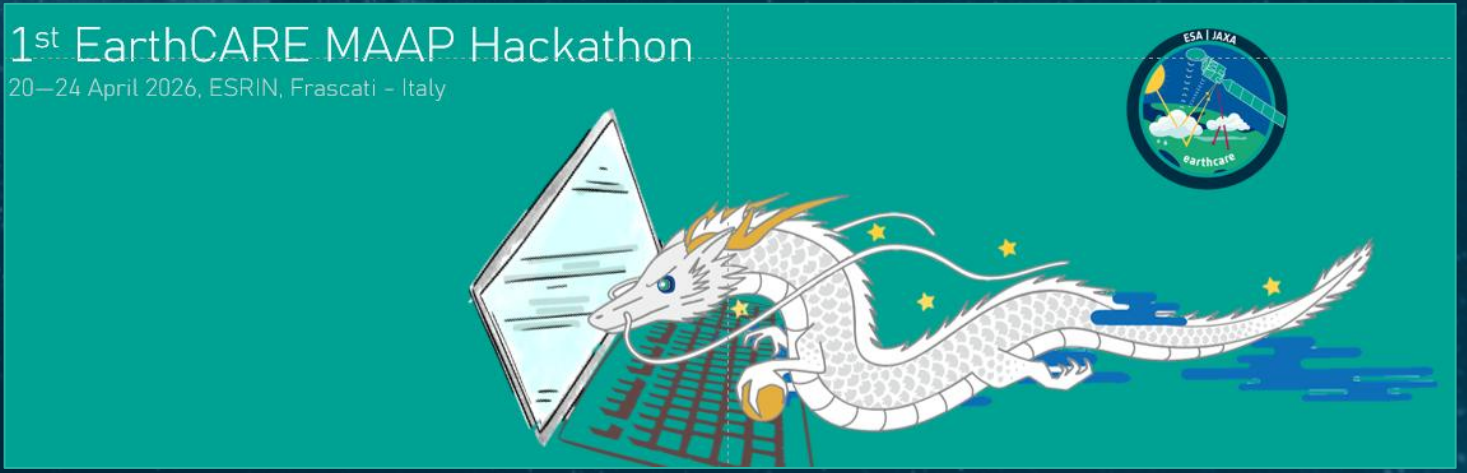
EarthCARE MAAP Hackathon Meet-and-Greet

Saskia Brose and Beniamino Abis, 12/03/2026



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Important Information

EarthCARE Mission Intro Lecture on March 27*, 13.00 – 14.00 CET

Join: https://teams.microsoft.com/meet/38944596154687?p=YCz3w0etQd4k9iUA4M

Meeting ID: 389 445 961 546 87 ; Passcode: 6iy6S5aw

* Will be recorded

MAAP Setup & Introduction on March 31*, 13.00 – 14.00 CEST

Join: https://teams.microsoft.com/meet/32869151705726?p=X8nFvJTCzdCF4qdEtv

Meeting ID: 328 691 517 057 26 ; Passcode: rX7KL3qk

Slack: earthcare-hackathon.slack.com

GitHub: smbrose/earthcare-maap-hackathon

Timeline

Today

Meet-and-Greet

March 27th

EarthCARE Lecture*

March 31st

MAAP Introduction & Setup*

Monday – 20/4

Please arrive by 13.45

Welcome & Setup

Aperitivo, ESRIN

Tuesday – 21/4

Hack!

Guided Tours at ESRIN #1

Wednesday – 22/4

Hack!

Science Lecture

Thursday – 23/4

Hack!

Guided Tours at ESRIN #2

Hackathon Dinner, Frascati

Friday – 24/4

Final hacking...

Closing session & final presentations

* Will be recorded

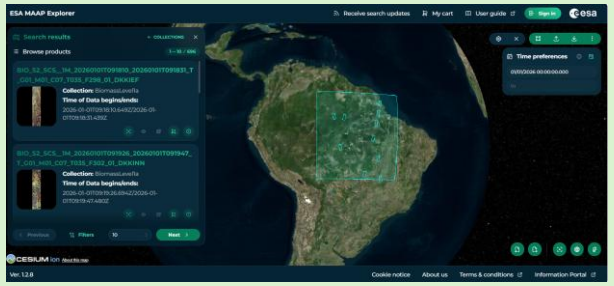
ESA MAAP

ESA MAAP
Data Access

Catalogue

OPENAPI STAC OpenSearch

Explorer - GUI



ESA MAAP
Collaborative Environment

CODING

PROCESSING

Algos/
Processors

Scalable
Resources

Other
Services

VISUALISATION &
ANALYTICS

ESA MAAP – Request account

Step 1: Create an EO Sign In Account

1. Visit: https://eoiam-idp.eo.esa.int/
2. Step-by-step instructions: https://earth.esa.int/eogateway/faq/how-can-i-create-an-eo-sign-in-account-for-non-esa-eo-data-users-

Step 2: Join the EarthCARE MAAP Collaborative Platform

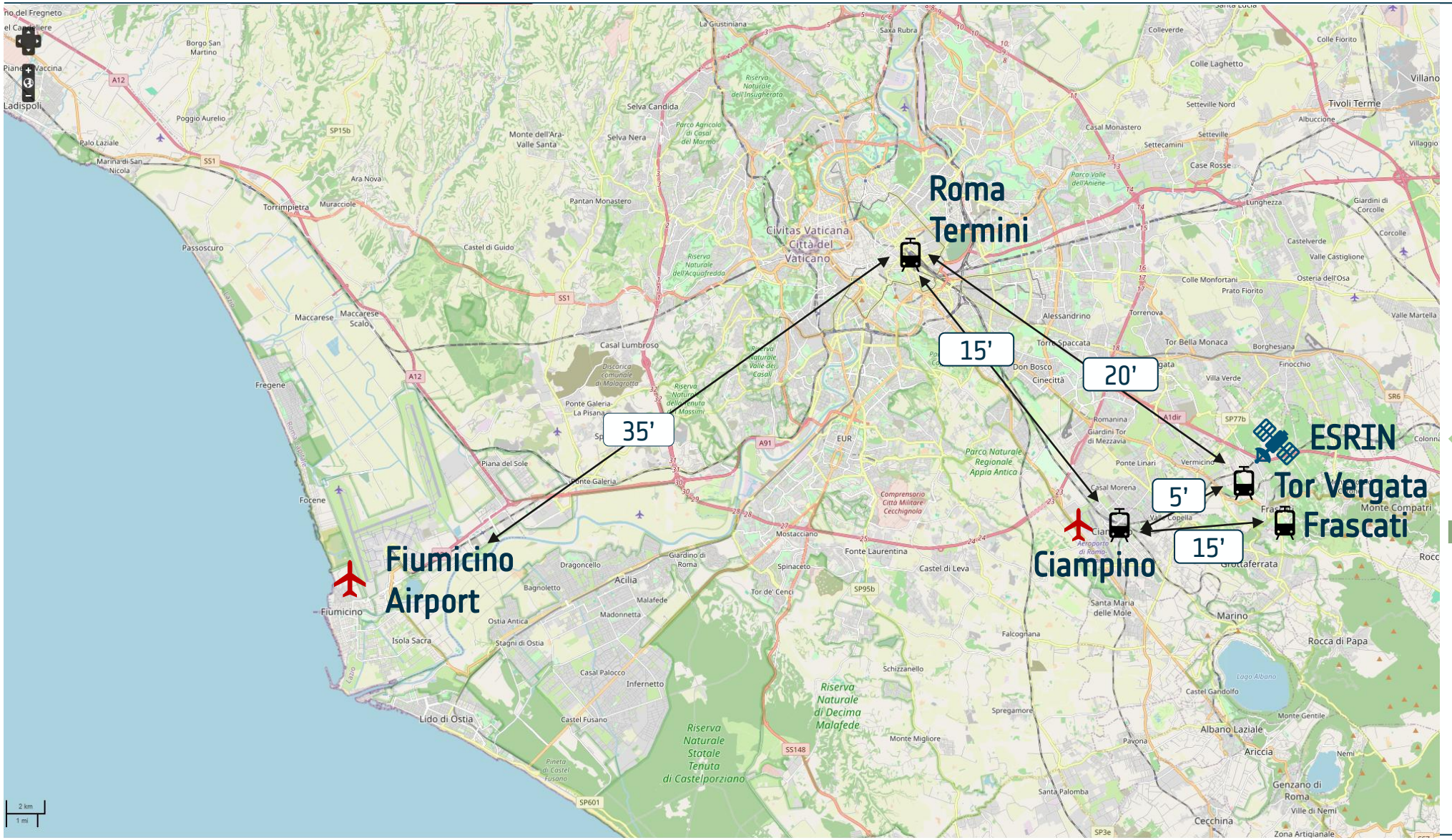
1. Access the platform via the link provided: Join Collaborative Environment – EarthCARE
2. **Important:** Please **specify that you are participating in the hackathon** to expedite approval.

All the links and information will be available on GitHub

ESA MAAP
Data Access

ESA MAAP
Collaborative Environment

Where to stay & getting to ESRIN



Train Station

Airport

ESRIN

ESRIN Site Map

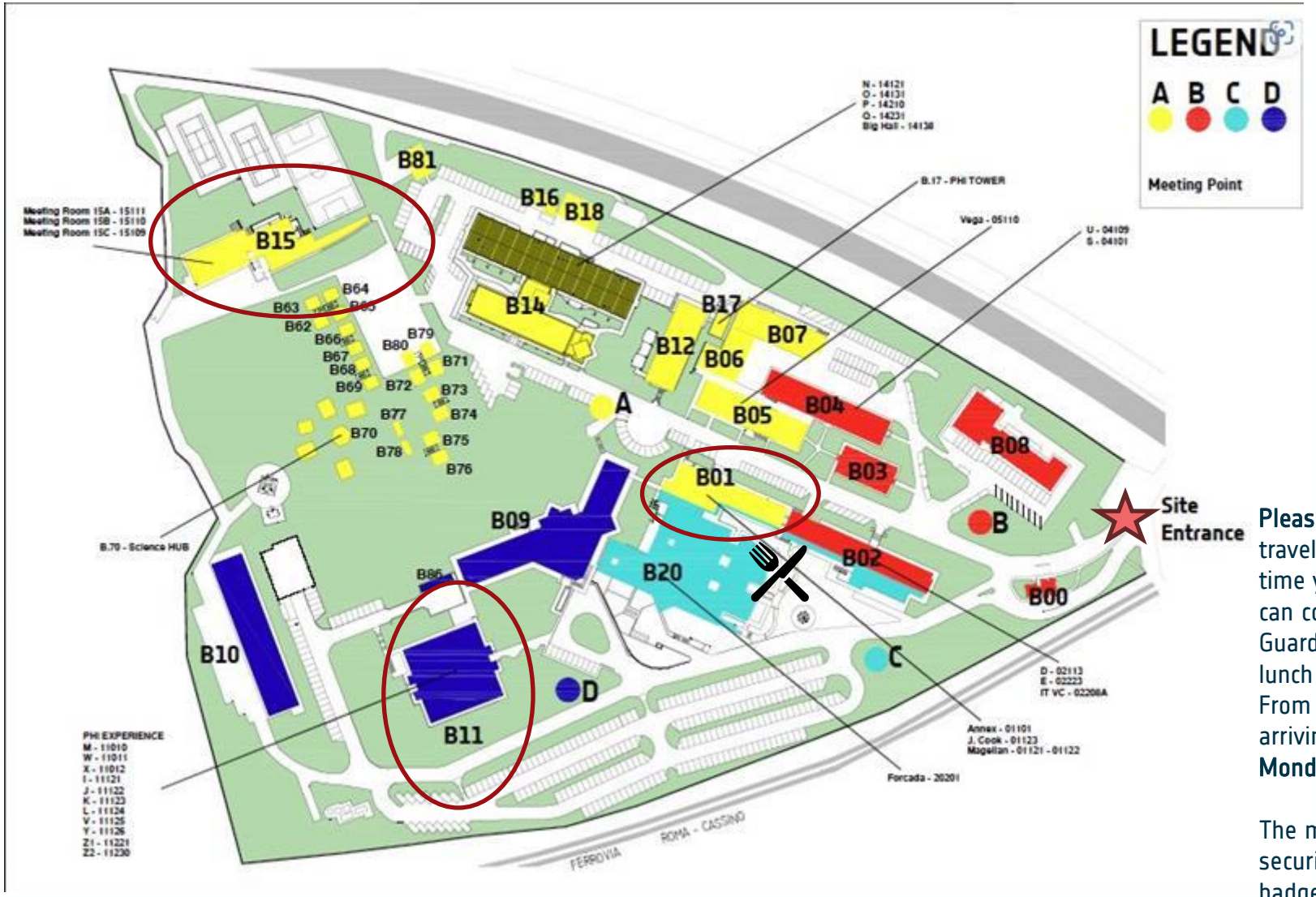
Site Entrance

B15 – Hack rooms

B11 – ESRIN Tour

B01 – Magellan (Opening, Closing Sessions)

Cantine / Bar



Please note: Once you book your travels, you can let us know at what time you plan to arrive at ESRIN so we can coordinate with the ESRIN Security Guards. You are also welcome to have lunch at the canteen on-site.
From Roma Termini there is a train arriving at **12:32** and at **13:32 on Monday April 20th at Tor Vergata.**

The map will also be handed to you by security together with your temporary badge by security.

Travel Support

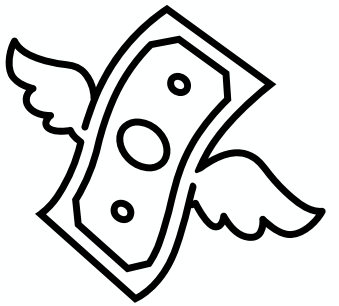
Budget for the event is **limited**

Reimbursement of expenses up to around **400 euro per participant for selected participants**

Precedence will be given to (graduate/PhD) **students that have no other funding available**

Keep all the receipts and boarding passes to submit them at a later point

More detailed instructions and confirmation of support will follow once we consolidate the list of participants



Topics

Cloud and Cloud-Aerosol Interactions — 9

- Stratocumulus Clouds:
  - Cloud structure transitions
  - Aerosol hitting clouds
  - Below cloud aerosol
- Monsoonal precipitation

Machine Learning versus EarthCARE — 5

- Classifications
- 3D reconstruction
- Detections

Comparisons — 10

- PACE, CALIPSO
- ECVT, CalVal, In-Situ Activities
- Climate Models, Regridding

Global Maps (Monthly / Seasonal) — 7

- E.g. Global Snowfall Estimates

Visualisations & Data Management — 8

- Creation of visualisations, plots, dashboard/infographics
- Different data formats (h5 vs zarr)

Detection — 6

- Ship tracks
- Contrails
- Ashes and SO_2
- Haboob/Others

Lightning — 5

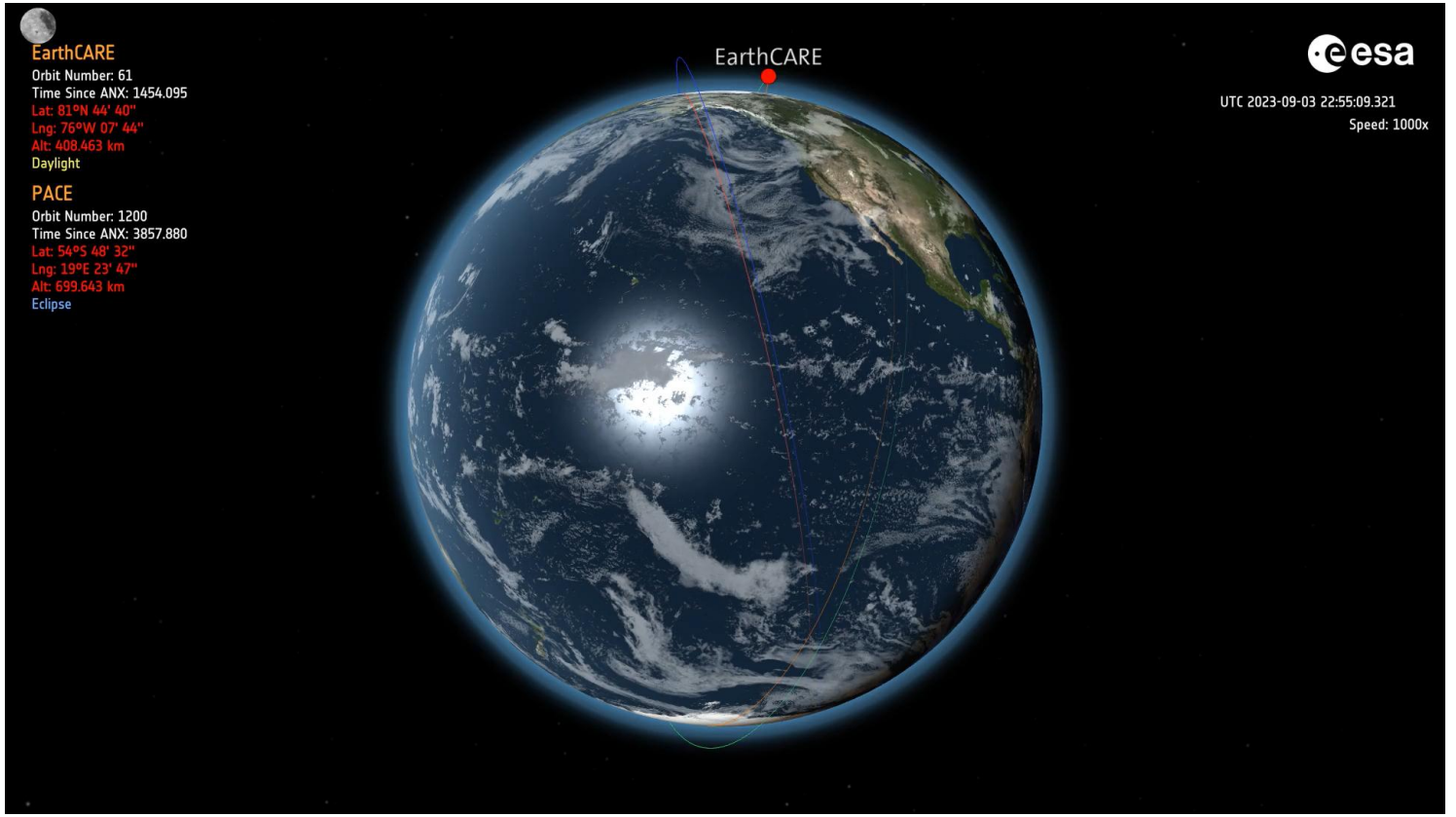
- Vertical structure of storms
- Prediction during convective storms

n # of participants with initial interest

Remote PACE Node

Plankton, Aerosol, Cloud, ocean Ecosystem mission

8 OEL'ers (NASA Ocean Ecology Laboratory) participating remotely through the *PACE node* of the EarthCARE MAAP Hackathon -
> PACE-EarthCARE Intercomparison



By NASA - https://fpd.gsfc.nasa.gov/images/missions/PACE.png, Public Domain, https://commons.wikimedia.org/w/index.php?curid=145047779

Who are you?

Institution / Current position

Experience with atmospheric sciences and/or the EarthCARE mission

Would you be interested in being an expert in your hack team?

What would you like to work on?